

Compatibility Gap Analysis in 10-Minute Home Help Platforms: Challenges, User Frictions, and a Compatibility-Aware Matching Framework

Independent Research Study

Pain Points • Research Findings • Proposed Solutions • Workflow

June 2026

Prepared by:

Rishita Rathi

B.Tech, Department of Artificial Intelligence & Data Science
Thakur College of Engineering and Technology, Mumbai, India

Email: itsrishitarathi345@gmail.com

TABLE OF CONTENTS

A. Abstract	1
B. Introduction	1
1. OBJECTIVES OF THE STUDY.....	2
2. LITERATURE SURVEY	3
3. Origin of This Study.....	4
4. How These Apps Currently Work?.....	4
5. Identified Pain Points.....	4
6. Core Research Insight.....	7
7.The Supply-Demand Conflict	7
8. The Househelp Onboarding Challenge	8
9. The AI Voice Assistant — Proposed Solution	9
10. Proposed End-to-End Workflow	11
11. Summary of Findings & Recommendations.....	16
12.References	16

A. ABSTRACT

The rapid growth of on-demand home service platforms has transformed the way households access cleaning and domestic assistance. Applications such as Pronto and Snabbit promise quick service fulfillment by matching users with available househelps within a short time frame. However, existing matching systems primarily focus on availability and proximity, often overlooking critical compatibility factors between the user and the assigned househelp. These factors include health conditions, allergies, religious beliefs, cultural practices, language preferences, safety concerns, physical limitations, and familiarity with specific tools or household requirements.

This research investigates the compatibility gaps present in current 10-minute home help applications and analyzes how these gaps contribute to service inefficiencies, user dissatisfaction, and operational challenges. Through a detailed examination of real-world friction points, the study identifies six major categories of compatibility mismatches and highlights the limitations of availability-based matching models.

To address these challenges, the research proposes a dual-profile compatibility framework in which both users and househelps maintain structured

compatibility profiles. Additionally, an AI-powered voice onboarding system is introduced to enable inclusive participation of househelps who may have limited digital literacy. The proposed solution aims to improve service quality, reduce doorstep conflicts, increase transparency during booking, and establish a trust-driven ecosystem for home services. The findings suggest that compatibility-aware matching can serve as a significant competitive advantage for home service platforms while improving satisfaction and retention for both users and service providers.

Keywords: Home Service Platforms, Compatibility Matching, AI Voice Assistant, User Experience, Househelp Onboarding, Service Personalization, Human-Centered Design, Pronto, Snabbit, On-Demand Services

B. INTRODUCTION

The emergence of digital platforms has significantly transformed service delivery across multiple industries, including transportation, food delivery, healthcare, and domestic services. In recent years, on-demand home help applications such as Pronto and Snabbit have gained popularity by offering users quick access to household assistance within minutes. These platforms leverage technology to streamline booking, scheduling, and dispatch processes, creating a convenient experience for users while

generating employment opportunities for househelps.

Despite these advancements, the current matching mechanism employed by most home service platforms remains heavily dependent on helper availability. Once a user selects a service and preferred time slot, the system assigns the nearest available househelp without considering whether the assigned individual is personally compatible with the household environment. While this model enables rapid service delivery, it often results in unforeseen challenges that become apparent only after the helper arrives at the user's residence.

Compatibility in home services extends beyond professional skills and task execution. Households differ significantly in terms of cultural practices, dietary preferences, presence of pets, safety requirements, language preferences, physical environment, and privacy expectations. Similarly, househelps may possess personal limitations, health conditions, religious restrictions, physical constraints, or communication barriers that affect their ability to comfortably perform certain tasks. When these factors are not considered during the matching process, conflicts, dissatisfaction, and service disruptions become more likely.

This study seeks to examine the compatibility challenges present in current 10-minute home help platforms and evaluate their impact on both users and service providers. The research identifies key friction points within existing workflows and proposes a compatibility-driven

matching framework designed to improve transparency, service quality, and overall user experience. Furthermore, the study introduces an AI-powered voice onboarding solution aimed at addressing the digital literacy challenges faced by a significant portion of the househelp workforce in India.

By shifting compatibility assessment from the doorstep to the booking stage, this research aims to establish a more efficient, inclusive, and trust-oriented model for home service delivery. The proposed framework not only enhances operational effectiveness but also contributes to a more human-centered approach to service matching in the growing home services industry.

1. OBJECTIVES OF THE STUDY

The primary objectives of this research are:

- To analyze the existing matching mechanisms used by 10-minute home help platforms and identify their limitations.
- To investigate compatibility-related challenges faced by both users and househelps during service delivery.
- To categorize major mismatch factors, including health, cultural, safety, language, physical, and operational constraints.
- To evaluate the impact of availability-based matching on service quality, customer satisfaction, and helper experience.

- To design a dual-profile compatibility framework that enables more informed and transparent matching decisions.
- To propose a classification system that distinguishes between hard requirements and soft preferences for both users and househelps.
- To develop an AI-powered voice onboarding approach that accommodates househelps with limited digital literacy and diverse language preferences.
- To create an end-to-end compatibility-aware workflow that integrates onboarding, matching, booking, and feedback processes.
- To explore strategies for balancing service speed with compatibility in high-demand operational environments.
- To provide recommendations that can help home service platforms reduce conflicts, improve trust, and achieve long-term user and helper retention.

2. LITERATURE SURVEY

Recent studies and industry reports have highlighted several challenges associated with platform-based home service applications. While platforms such as Pronto and Snabbitt have improved convenience through rapid service delivery, existing matching systems primarily focus on availability and location rather than compatibility between users and househelps.

The Week [1] reported that home-help platforms involve a higher level of trust and

personal interaction than delivery or ride-hailing services because workers spend extended periods inside users' homes. This increases the importance of ensuring compatibility and safety during service allocation.

According to a YourStory report [2], Pronto's matching process is largely based on geographical proximity and helper availability to maintain its 10-minute service promise. However, compatibility factors such as personal preferences, cultural considerations, and physical constraints are not considered during matching.

Research conducted by the Association for Progressive Communications (APC) [3] found that domestic workers often face information asymmetry and possess limited digital literacy, making it difficult for them to independently manage platform-related processes. Similarly, the Centre for Internet and Society (CIS) [4] observed that many domestic workers rely on basic mobile devices and require simple, accessible onboarding mechanisms.

Studies on domestic work platforms [5] further indicate that worker preferences and constraints are often not adequately captured by existing platform designs, resulting in mismatches and reduced satisfaction. To address accessibility challenges, NITI Aayog's Frontier Tech initiative [6] demonstrated the effectiveness of voice-first AI systems for engaging workers with varying literacy levels. Additionally, People+AI [7] highlighted the growing role of multilingual voice technologies in bridging India's digital divide.

Research Gap

Existing literature discusses platform efficiency, worker welfare, digital accessibility, and voice-based technologies. However, limited research focuses on compatibility-aware matching in home-help platforms. Furthermore, no study was found that combines dual-sided compatibility profiling with AI-powered voice onboarding for domestic workers. This research aims to address this gap by proposing a compatibility-driven matching framework for 10-minute home-help applications.

3. Origin of This Study

This research originated from a direct observation during usage of two popular 10-minute home help booking apps — Pronto and Snabbit. Both platforms allow end users to book a househelp for a specific time slot, with the primary matching criteria being availability: whoever is free gets dispatched.

The core research question that emerged was:

Research Question: What happens when the househelp who arrives at the user's home is fundamentally incompatible with the household's needs — not because of skill, but because of personal, health, cultural, or physical constraints that neither party disclosed at the time of booking?

This study maps all such compatibility gaps, proposes a dual-sided profile system to

resolve them, and addresses the unique operational challenge of onboarding a demographic of helpers who may not be technology-literate.

4. How These Apps Currently Work

Both Pronto and Snabbit follow the same core booking logic:

- User opens app and selects a time slot
- App checks which househelps are free in that slot
- A free househelp is assigned and dispatched
- User gets a confirmation with the helper's name and ETA

Critical Gap: At no point in this flow does the system check whether the assigned househelp is personally compatible with the user's household. The only variable is availability.

The result: mismatches are discovered at the doorstep — when the helper has already arrived and both parties are in an uncomfortable, unresolved situation.

5. Identified Pain Points

Through research and analysis, the following compatibility friction categories were identified. Each represents a real scenario where the current availability-only matching fails both the user and the househelp

5.1 Health & Allergies

Category	User discloses	Househelp discloses
Health & Allergies	• Pet in the household (dog, cat, bird) • Non-vegetarian household with raw meat • Specific cleaning chemicals in use • Heavy dust or mold in old furniture	• Allergy or fear of pets • Cannot handle non-veg utensils (religious or personal) • Skin sensitivity to certain chemicals • Respiratory issues triggered by dust

5.3 Safety & Privacy

Category	User discloses	Househelp discloses
Safety & Privacy	• Elderly, infant, or differently-abled person at home • Restricted rooms (home office, study with documents) • Security cameras present inside the home • Valuable items or unlocked spaces	• Not comfortable working under camera surveillance • No training or experience handling infants or elderly • Fear of confined or locked spaces • Discomfort with being alone in a home with certain individuals

5.2 Religious & Cultural Boundaries

Category	User discloses	Househelp discloses
Religious & Cultural	• Religious idols or pooja room requiring special handling • Alcohol or pork present in the kitchen • Separate utensils for religious reasons	• Cannot touch certain objects due to faith • Observes fasting days with limited physical capacity • Preference for household gender (e.g. solo female user)

5.4 Physical Ability & Task Limits

Category	User discloses	Househelp discloses
Physical Ability	• Heavy furniture needs to be moved • High ceiling fans, lofts, or exterior windows to clean • Specific appliances — chimney, oven,	• Back issues — cannot lift heavy items • Fear of heights or cannot work on ladders/stools • Not trained for specific appliances

	washing machine • Large floor area requiring extended effort	• Physical stamina limitations
--	---	--------------------------------

5.5 Language & Communication

Category	User discloses	Househelp discloses
Language	• Preferred language to give instructions • Relies on written task lists (sticky notes, checklists) • Complex verbal instructions during the session	• Languages understood and spoken • Literacy level — can or cannot read written instructions • Preferred mode of receiving instructions

5.6 Tools, Products & Supplies

Category	User discloses	Househelp discloses
Tools & Products	• Uses only specific branded cleaning products • No supplies at home — helper must carry everything • Fragrance-free or eco-friendly products only	• Only familiar with specific tools or equipment • Cannot carry own supplies due to transit constraints • Unfamiliar with certain appliance brands

5.7 Additional Friction Points

- Timing sensitivity — newborn napping, guest arriving soon; helper unaware of time-critical constraints
- Entry access confusion — OTP, hidden key, building security check-in not communicated in advance
- Task scope creep — user adds tasks during the visit that were never disclosed at booking time
- Helper substitution — a different helper arrives than the one whose profile may have been seen
- No post-service compatibility verification — no mutual flag for mismatch for future matching

6. Core Research Insight

Key Insight: Both Pronto and Snabbit currently optimise for availability. They treat househelps like taxis — whoever is closest and free gets the job. This works for rides. It fails for home services, where personal compatibility between the helper and the household is a fundamental requirement, not an afterthought.

The mismatch is not about skill. Both platforms do a reasonable job of ensuring helpers are trained for standard cleaning tasks. The mismatch is about personal, cultural, health-related, and physical compatibility — dimensions that cannot be inferred from availability data alone.

The insight that anchors the entire proposed solution is this:

Design Principle: Move the discovery of incompatibility from the doorstep — where nothing can be done — to before booking, where both sides can make an informed choice.

7. The Supply-Demand Conflict

A critical operational challenge was identified during research: what happens when a user's compatibility requirements are met by a helper who is NOT available in the selected slot, while the only available helper has a known mismatch?

This is the supply-demand conflict. Currently, the app resolves it silently by dispatching whoever is free. The proposed solutions below address this conflict explicitly.

7.1 Option 1 — Compatibility-First Matching

Show the user only helpers who both match their profile AND are free. If no compatible helper is available right now, show the next available time slot where a compatible helper exists. The platform's 10-minute promise evolves to: '10 minutes for a matched helper.'

7.2 Option 2 — Tiered Matching with Transparency

Present two sections to the user at booking:

- Best match available now (may be partial compatibility)
- Fully compatible helper available at X time

Let the user choose between speed and fit. This is honest and gives the user agency.

7.3 Option 3 — Partial Match with Disclosure

If the only available helper has a known soft mismatch, the app flags it before confirmation:

Example disclosure: This helper has indicated discomfort with non-veg utensils. Do you still want to proceed?
Yes / No / Find next available match

The user decides — not the system. This is the most respectful design for edge cases.

7.4 Option 4 — User-Side Flexibility Flags

During profile setup, ask the user to classify each preference:

Hard requirement	Soft preference
Cannot be overridden — block incompatible helpers	Show a warning but allow booking
e.g. severe pet allergy, religious constraint	e.g. mild preference for female helper
Filters the helper pool entirely	Allows user to consciously accept the mismatch

7.5 Option 5 — Supply-Side Fix (Long-Term)

As the platform scales, ensure every operational zone has helpers with diverse compatibility profiles. Gaps in compatibility become a hiring and onboarding challenge, not a product problem. This is a medium-to-long-term fix that complements the above product solutions.

Recommended Approach: Implement Options 2 + 3 + 4 together. Option 2 gives the user a clear choice. Option 3 handles edge cases transparently. Option 4 lets users control what is non-negotiable vs. flexible. Option 5 is pursued in parallel as a supply operations goal.

8. The Househelp Onboarding Challenge

A critical question was raised during research: how does a househelp fill her compatibility profile if she is not tech-savvy, not smartphone-comfortable, or has limited literacy?

This is the last-mile onboarding problem. The helper is the product — but she cannot be expected to onboard herself digitally. The platform must own this responsibility.

8.1 The Reality of the Househelp Demographic in India

- Feature phone or low-RAM Android, often shared with family
- Semi-literate or literate only in a regional language
- Not comfortable with forms, dropdowns, or digital toggles
- Data pack anxiety — will not spend time on a heavy app
- Often registered via a contractor or aggregator, not directly

8.2 Who Does the Onboarding?

Three realistic actors can own helper onboarding:

Actor 1: Company Field Agent (Most Reliable) When a helper is recruited — walk-in, referral, or via contractor — a company agent sits with her and fills the preference profile through an agent-facing admin panel. The agent asks questions verbally in her language, selects the

answers, and submits. The helper never touches a form.

Actor 2: Contractor / Aggregator

Many helpers come through a middleman. The platform makes the contractor responsible for submitting profiles for every helper they supply. Contractors already know their workers' basic constraints — the profile system formalises this knowledge.

Actor 3: AI Voice Assistant (Breakthrough Solution)

The most effective and scalable solution identified in this research: a voice-based onboarding assistant in the helper's preferred regional language.

9. The AI Voice Assistant — Proposed Solution

The AI voice assistant removes every barrier at once: literacy, tech comfort, language, and form fatigue. It is the strongest UX solution for helper onboarding and the most significant product innovation proposed in this study.

9.1 The Experience for the Helper

The helper opens the app — or simply opens WhatsApp, which she already has. She taps one button labelled 'Start'. An AI voice begins speaking in her selected language. Language selection is done via flag icons or language name cards — no reading required.

The voice asks simple, conversational questions — not form language, but actual spoken language:

Example (Marathi): "Tumhala kadhihi kutrya kinva manjarishI kaam karayla tras hotil ka?" (Do you ever have trouble working in homes with dogs or cats?)

She responds by tapping one of two or three large buttons on screen, no scrolling, no reading required. The responses are automatically stored and mapped to the helper's compatibility profile.

9.2 Handling Nuance — Three-Option Model

Some preferences are not binary. The voice assistant uses a three-option model to capture real-world nuance:

Ho, karti	Thoda problem	Nahi jamnar
Yes, I can do it	Somewhat uncomfortable (soft flag)	No, I cannot (hard block)

The middle option becomes a soft flag in the matching system — a warning shown to the user — rather than a hard block. This preserves booking flexibility while maintaining transparency.

9.3 Profile Built Over Time — Not All at Once

The helper's profile is built in three layers across her tenure on the platform:

- At recruitment — hard constraints collected by agent or voice assistant (allergies, religious limits, physical limits). These are deal-breakers.
- After first few jobs — soft preferences emerge from ratings and feedback. If a user flags that the helper was uncomfortable with pets, the system auto-suggests a profile update.
- Periodic re-verification — every 3 months, the voice assistant re-engages her to check if anything has changed.

9.4 Technology Stack

Component	Solution
Speech synthesis in Indian languages	Sarvam AI (purpose-built for Indian languages), Google TTS, or Microsoft Azure
Voice recognition (optional)	Tap-only is safer and more reliable; spoken input is an enhancement
Delivery channel	WhatsApp Business API — no app download required
Backend logic	Simple state machine mapping voice responses to profile fields

Re-engagement	Automated WhatsApp message every 3 months for profile refresh
---------------	---

9.5 Why This Becomes a Competitive Moat

Strategic Insight: Word spreads fast in helper networks. If one helper tells another 'Ek app hai jo Marathi mein baat karta hai aur poochta hai ki mujhe kya nahi karna' — that becomes the platform's most powerful recruiting tool. It signals that the platform respects the helper as a stakeholder, not just a resource to be dispatched.

10. Proposed End-to-End Workflow

The following workflow describes how the dual-profile compatibility system integrates into the existing booking flow of both platforms.

10.1 User Onboarding Flow

- User downloads app and registers
 - Standard signup: name, address, phone
- Compatibility profile setup (one-time, editable later)
 - Pets at home — type and count
 - Dietary household type — veg / non-veg / both
 - Religious or cultural notes

- Vulnerable persons at home — infant, elderly, disabled
- Restricted areas in the home
- Heavy tasks expected — furniture, heights, appliances
- Preferred language for communication
- Cleaning products available vs. helper must bring
- Profile saved — used silently for all future bookings
- Filters out helpers with hard-block conflicts
- Ranks remaining available helpers by compatibility score
- Flags soft-preference mismatches for disclosure
- User sees tiered results:
 - Best compatible match available now
 - Fully compatible match available at [time]
- If partial match only: disclosure shown before confirmation
- User confirms booking with full awareness of any flags
- Helper receives job notification with household context summary
 - Pets present / no pets
 - Non-veg household / veg household
 - Any special notes relevant to her profile
- Post-service: both sides rate compatibility, not just task quality
- Mismatch flags auto-update profiles for future accuracy

10.2 Househelp Onboarding Flow

- Helper recruited via field agent, contractor, or walk-in
- Agent or contractor initiates voice assistant session
 - Language selected via icon — no reading required
 - AI voice asks questions conversationally in chosen language
 - Helper taps Yes / Somewhat / No buttons
 - All responses mapped to profile fields in backend
- Hard constraints stored as blocking flags
- Soft preferences stored as warning flags
- Profile verified and activated by operations team
- Re-engagement voice session every 3 months

10.3 Booking & Matching Flow

- User selects task type and time slot
- Matching engine runs compatibility check:

10.4 The Key Design Principle

Burden of Technology: The burden of technology must fall on the side that can carry it. The user carries the app experience. The company carries the helper's digital presence on her behalf. The matching engine is invisible to both — it simply ensures that when the doorbell rings, both sides are prepared for each other.

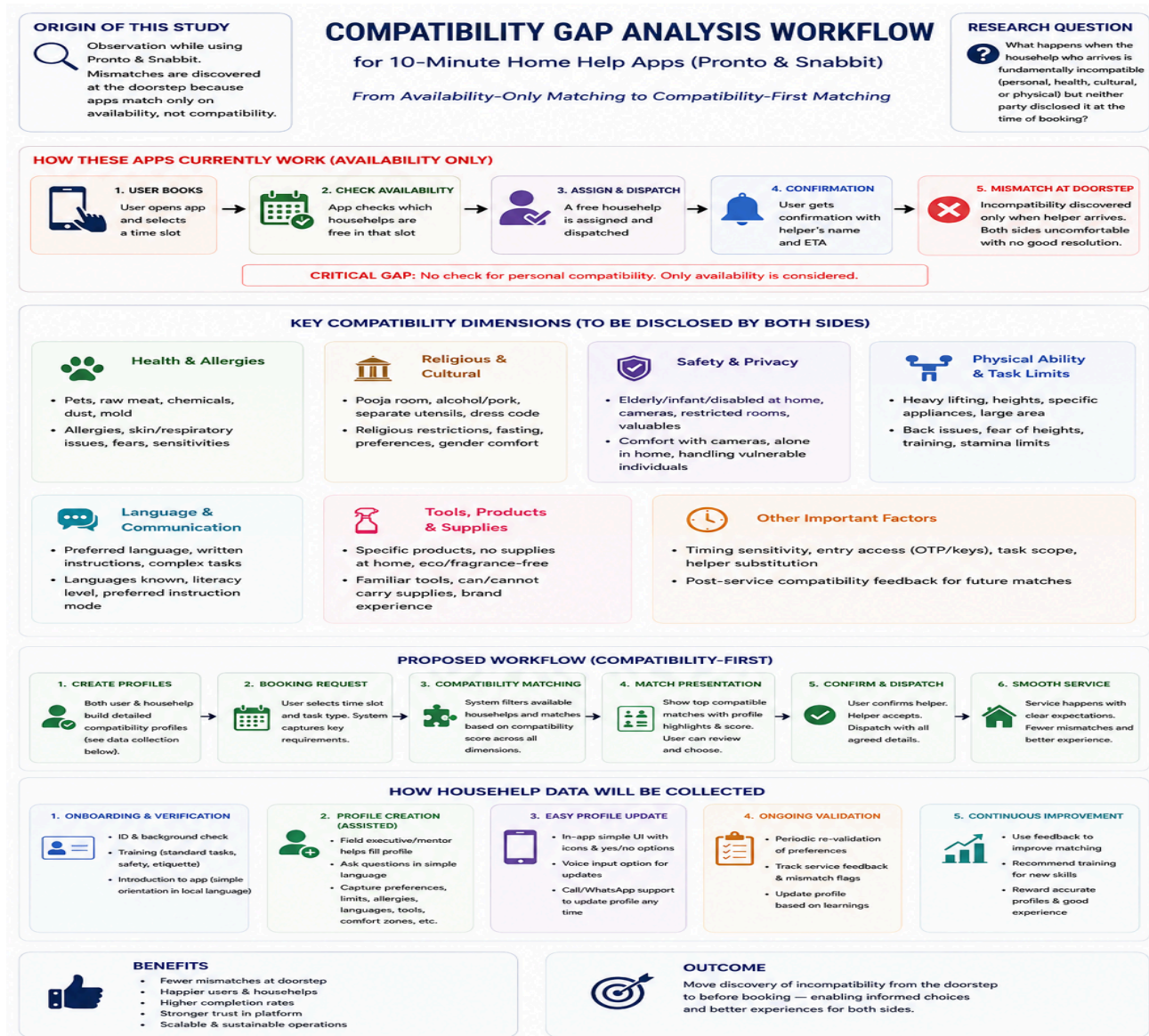


Fig. 1. Compatibility Gap Analysis Workflow.

10.5 WORKFLOW with respect to both the ends:

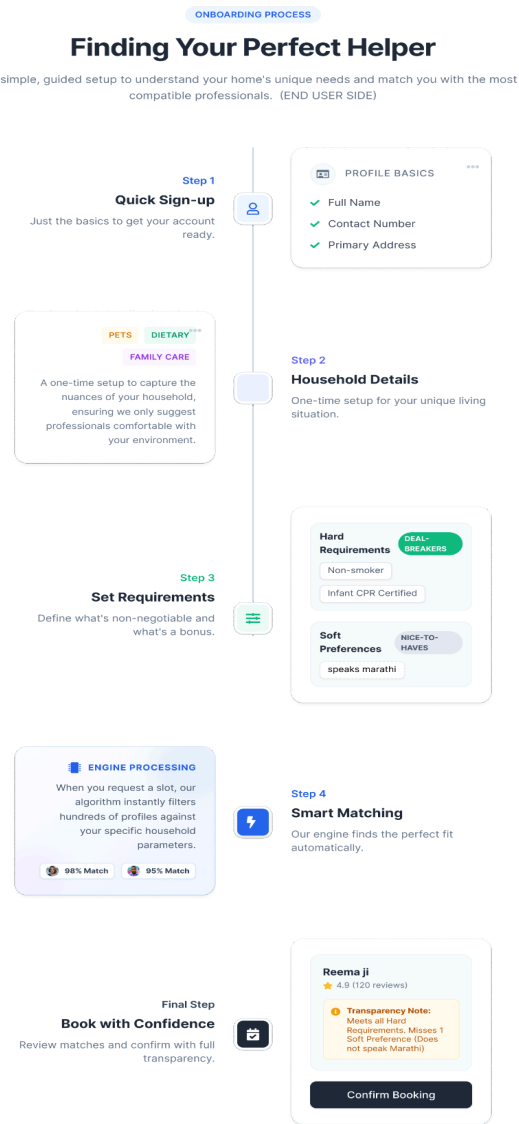
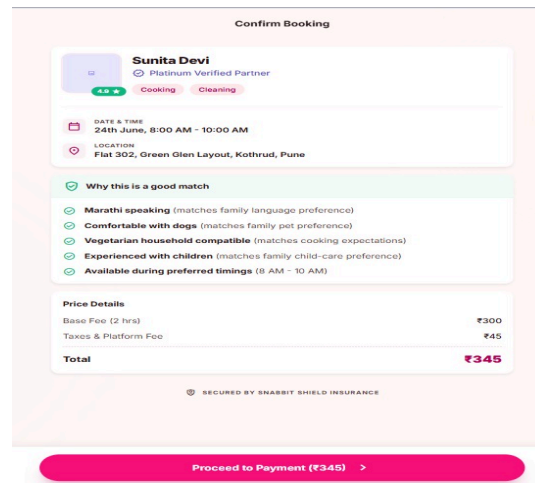
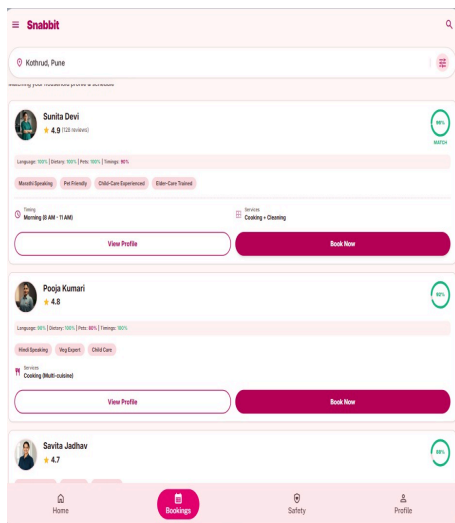
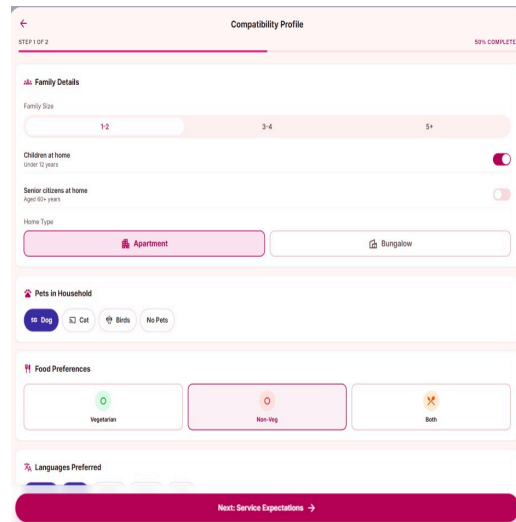
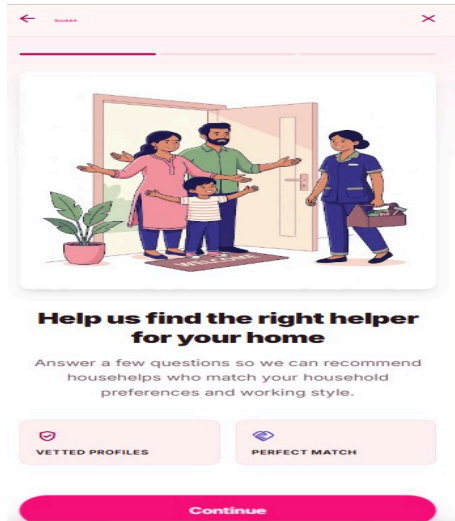


Fig 2: End User workflow



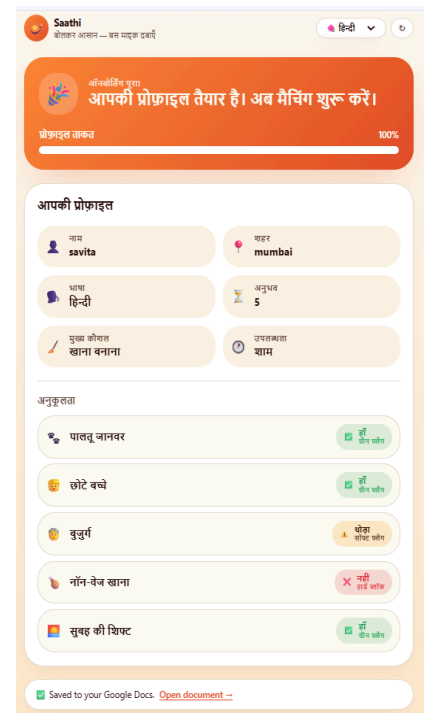
Fig 3: Househelp professional workflow

10.6. Prototype Screens – User Interface Mockups (Designed using Figma)





(Developed using Lovable)



11. Summary of Findings & Recommendations

Pain Points Identified

- No compatibility profile exists for either users or helpers on Pronto or Snabbit
- Matching is done purely on availability — incompatibility discovered at doorstep
- 6 major categories of mismatch: health, cultural, safety, physical, language, tools
- Supply-demand conflict: compatible helper may not be free at the required slot
- Househelp demographic cannot self-onboard digitally
- No post-service compatibility feedback loop exists

Recommended Solutions

- Dual-sided compatibility profile — user fills digitally, helper onboarded via AI voice assistant
- Tiered booking UI — show speed option vs. compatibility option to the user
- Hard vs. soft preference tagging — user controls what is non-negotiable
- Pre-confirmation mismatch disclosure — transparency before the booking is confirmed
- Post-service compatibility rating — separate from task quality rating
- Periodic profile refresh — automated voice check-in every 3 months for helpers

- Supply operations goal — build helper pools with diverse compatibility profiles per zone

The Opportunity

Competitive Advantage: Whichever platform — Pronto or Snabbit — builds this dual-profile compatibility system first will create a structural advantage that is very difficult to copy quickly. The helper profiles become richer and more accurate over time, matches improve, doorstep conflicts reduce, and retention improves on both sides. It is not just a UX feature — it is a trust infrastructure for the home services category.

Limitations of the Study

This study is based on secondary research, platform analysis, workflow evaluation, and problem-framing observations. The proposed framework has not yet been validated through live deployment or large-scale user testing and should be considered conceptual solution for further exploration and implementation.

12. REFERENCES

- [1] The Week, “Pronto, Urban Company, Snabbitt: India's Home-Help Conundrum,” Apr. 2026. [Online]. Available: <https://www.theweek.com>
- [2] YourStory, “Pronto's 10-Minute Model is Reshaping Domestic Help in Urban India,” Jan. 2026. [Online]. Available: <https://yourstory.com>
- [3] Association for Progressive Communications (APC), “Designing Domestic Work Platforms: A Case Study of Urban Company,” 2022. [Online]. Available: <https://www.apc.org>
- [4] Centre for Internet and Society (CIS), “Gender and Gig Work: Perspectives from Domestic Work in India,” 2021. [Online]. Available: <https://cis-india.org>
- [5] FIRN GenderIT, “Designing Domestic Work Platforms: A Case Study of Urban Company.” [Online]. Available: <https://firn.genderit.org>
- [6] NITI Aayog Frontier Tech Initiative, “Voice AI at Scale: Powering Half a Million Jobs a Year in India,” May 2026. [Online]. Available: <https://frontiertech.niti.gov.in>
- [7] People+AI, “Building India's Voice AI Future,” Aug. 2025. [Online]. Available: <https://peopleplus.ai>